RAJU BANDAM

B.Tech — Computer Science and Engineering (AIML)
Malla Reddy University, Hyderabad

+91 9121455643
rajubandam694@gmail.com
rajureddi rajureddie rajureddie.site

PROFESSIONAL SUMMARY

Motivated Computer Science and Engineering student specializing in AI/ML with strong hands-on experience in building real-time intelligent systems, full-stack applications. Skilled in developing end-to-end solutions Experienced in Python, Java, React, Flask, and deep learning, ML, with a strong command of data structures, algorithms, and cloud deployment.

EDUCATION

- **B.Tech in Computer Science and Engineering (AIML)** *2022–2026*
  Malla Reddy University CGPA: 8.6

- **Intermediate – MPC** *2020–2022*
  SR Junior College 90.1%

- **SSC** *2020*
  ZPSS T/EM CHENNUR CGPA: 10

PROJECTS

- **VisionAid++: OnDevice LLM for blind navigation and low vision enhancement (Github) IEEE PUBLICATION 2026**
  - Developed an Android-based assistive AI application that provides real-time blind navigation, low-vision enhancement, OCR, and voice guidance fully on-device for privacy and low latency.
  - Integrated **Vision-Language Models (VLLMs)** to analyze live camera feeds and generate safe navigation instructions such as obstacle alerts, distance awareness, and path guidance using Text-to-Speech.
  - Built multiple **low-vision accessibility modes** including glaucoma, macular degeneration, diabetic retinopathy, retinitis pigmentosa, high-contrast, night vision, and dynamic zoom enhancement.
  - Integrated OCR + NLP pipeline for reading text/signboards aloud.

  Tech Stack: Kotlin, Android, OpenCV, MNN, TensorFlow Lite, OCR, NLP, Hugging Face, Computer Vision.
- **AI-Based Smart Traffic Management System (GitHub) 2025**
  - Developed an adaptive traffic control system that dynamically optimizes green-light durations using real-time vehicle density analysis from multiple camera feeds.
  - Implemented emergency vehicle preemption to detect ambulances/fire trucks and automatically prioritize lane clearance during traffic cycles.
  - Built red-light violation detection using ANPR (EasyOCR) and a live dashboard with FastAPI, and TailwindCSS for monitoring traffic analytics.

  Tech Stack: Yolo, OpenCV, OCR, ANPR,React,FastAPI
- **BERT-based NER (G i t H u b) 2024**
  - Developed a **BERT-based Named Entity Recognition (NER)** system using **Python, Transformers, Hugging Face, and PyTorch** to identify entities such as names, locations, organizations, and dates from unstructured text. and OCR,
  - deployed on AWS EC2.
  - Designed React frontend for dynamic entity filtering and batch document processing.
  - TailwindCSS for monitoring traffic analytics.

  Tech Stack: Flask,Reactjs,AWS
- **Backlog Wallah — Learning Support App 2023**
  –Developed an Android based education-support app using Python + KivyMD library enabling access to syllabus, materials, and recorded classes for clearing Backlogs for Btech students.

CERTIFICATIONS

- NPTEL Certification — Java Programming
- Machine Learning Specialization: DeepLearning.AI
- AWS Cloud Practitioner
- Front-End Developer Certificate:Meta

TECHNICAL SKILLS

Languages: Python, Java, JavaScript **FrameWorks:** React, Flask, Node.js, FastAPI, KivyMD **ML/AI:** TensorFlow, PyTorch, Scikit-learn, OpenCV, HuggingFace, NLP, CV, Transformers
Tools: Git, GitHub, MySQL, REST APIs,AWS
Interests: Full-Stack Development, AI/ML Applications